APEX INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Experiment : 10

SUBJECT: Competitive Coding-1

SUBJECT CODE: 24CSP-291

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SECTION: 24BDS-1(B)

Aim:

1. The Celebrity Problem

►You are given an n × n matrix where mat[i][j] = 1 means person *i knows person j*.

► A **celebrity** is someone who is known by everyone but does not know anyone.

► Find the index of the celebrity, or return -1 if no celebrity exists.

Objectives:

- To understand **matrix-based relationship problems**
- To apply **elimination technique** for optimization
- To reduce time complexity from **$O(n^2)$ to $O(n)$**
- To practice **logical reasoning using conditions**
- To improve efficiency in **searching special elements**

Pseudocode:

1. The Celebrity Problem

```
FUNCTION celebrity(mat):

    n = number of people
     candidate =
0

    // Step 1: Find candidate
FOR i from 1 to n-1:         IF
mat[candidate][i] == 1:
            candidate = i

    // Step 2: Verify candidate
FOR i from 0 to n-1:
        IF i ≠ candidate:
            IF mat[candidate][i] == 1 OR mat[i][candidate] == 0:
                RETURN -1

    RETURN candidate
```

Code:

1.The Celebrity Problem

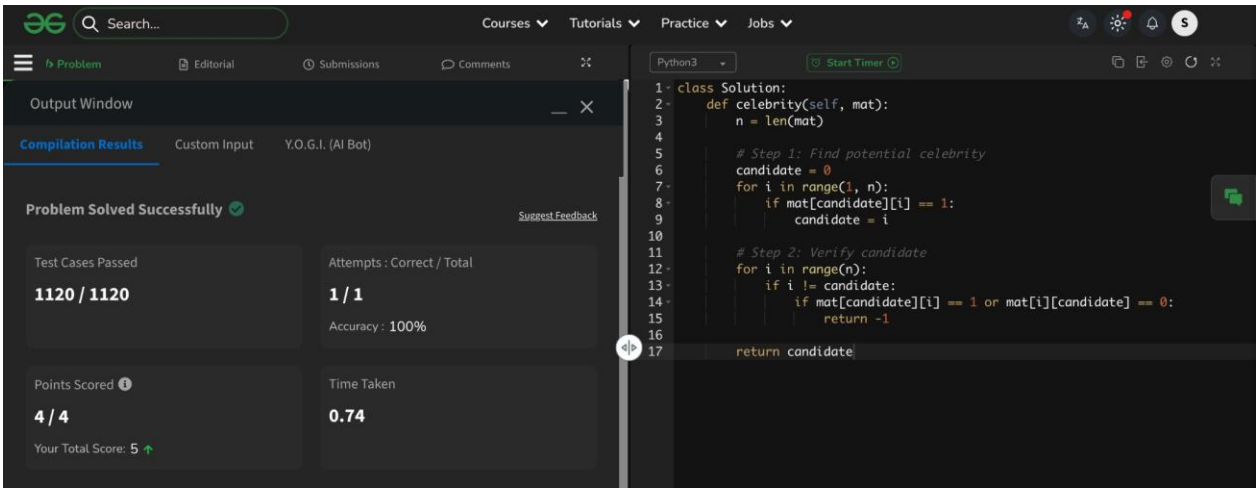
```
class Solution:     def
celebrity(self, mat):
        n = len(mat)

        # Step 1: Find potential
celebrity         candidate = 0
for i in range(1, n):             if
mat[candidate][i] == 1:
                candidate = i

        # Step 2: Verify candidate         for i in range(n):
if i != candidate:                 if mat[candidate][i] == 1 or
mat[i][candidate] == 0:
                    return -1

        return candidate
```

Output:



Learning Outcome:

- Identify a **celebrity using elimination strategy**
- Use **matrix representation for relationships**
- Optimize brute-force solutions into **O(n)**
- Perform **validation checks efficiently**
- Strengthen problem-solving for **interview puzzles** ● Apply logic to similar problems like:
    - Leader in array
    - Majority element